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# Leveraging machine learning for intelligent agriculture

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**Abstract**
Agriculture, the backbone of many economies, faces challenges like lack of information, outdated practices, and limited access to technology, hindering farmer productivity. This work proposes a user-friendly, multilingual platform leveraging Generative AI to address farmers' diverse needs. The platform encompasses various features to enhance agricultural practices. An LLM-powered Government Scheme Advisor functions as a multilingual chatbot offering intelligent guidance on government agricultural schemes and subsidies. The Disease Detection module utilizes AI technology for real-time identification and treatment recommendations, minimizing crop diseases and yield losses. The Soil Testing Centre feature locates nearby soil testing centers, providing essential information based on geographical data to assist farmers in optimizing soil quality. A Crop Recommendation feature employs Machine Learning algorithms to offer personalized crop recommendations, considering various factors and aiding informed decision-making. The Crop Planning Tool, with its intuitive user interface, simplifies planning planting schedules and managing resources. Additionally, the platform includes an MSP Center Locator to find nearby Minimum Support Price (MSP) centers based on location. By integrating these innovative solutions, this platform bridges the gap between conventional agricultural techniques and contemporary technology, equipping farmers with the resources and expertise essential for advancing productivity and sustainability. Multilingual support ensures accessibility for a wider audience, breaking down language barriers and promoting inclusivity in the agricultural sector. This work proposes an innovative, multilingual platform powered by Generative AI to address these issues. Key features include an LLM-driven chatbot for government scheme guidance, AI-based real-time disease detection, and location-based tools for soil testing and MSP center identification. Additionally, the platform offers personalized crop recommendations and intuitive crop planning tools, bridging the gap between traditional methods and modern technology. Multilingual support ensures inclusivity, empowering farmers with resources to enhance productivity and sustainability. Future research could focus on improving precision agriculture, climate resilience, market intelligence, and socio-economic assessments to further advance the agricultural sector.

**Article Highlights**

- A multilingual AI platform helps farmers improve crop management, health monitoring, and resource planning.
- Personalized guidance on government schemes empowers farmers with crucial support and funding opportunities.

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- The platform bridges traditional farming with modern AI, enhancing productivity and sustainability for all users.

**Keywords** Machine learning · Smart agriculture · Computer vision · LLM · Deep learning approaches

## 1 Introduction

Agriculture, the backbone of many nations, relies heavily on farmers' dedication, yet faces challenges like lack of information, outdated practices, and limited technology access. AI's introduction to agriculture offers novel solutions by providing precise, data-driven insights to improve decision-making and productivity. However, a significant research gap exists in developing comprehensive, user-friendly platforms that integrate various AI-driven tools to address the diverse challenges faced by farmers, particularly in management and centralized platform solutions [18].

Our solution addresses this research gap by proposing a comprehensive, multilingual platform leveraging Generative AI to empower farmers. Key focus areas include Crop Management, Crop Health, Access to Essential Information, and Efficient Planning, all centralized on one platform. The integration of these diverse tools into a single platform provides a solution that is both accessible and practical for farmers. This approach bridges the gap between traditional farming methods and modern technology, enhancing productivity and sustainability. Particularly novel is our LLM-powered Government Scheme Advisor, which offers personalized guidance on agricultural schemes, marking a significant innovation in how farmers interact with technology. Multilingual support ensures accessibility, fostering inclusivity in the agricultural community. This centralized approach not only enhances the management of resources but also introduces a new era of AI-driven solutions in agriculture.

The structure of the paper is as follows: the Literature Review is presented in Section II, Design and Implementation is covered in Section III, Results and Discussion are included in Section IV, and the Conclusion is provided in Section V.

## 2 Literature review

The literature on digital agriculture technologies in India highlights the potential of these technologies for sustainable agriculture, addressing the environmental, economic, and social dimensions. Precision Agriculture (PA) and Digital Agriculture (DA) are key approaches. PA focuses on optimizing productivity and sustainability using technology like satellite farming and Decision Support Systems. DA involves using sensors, IoT, remote sensing, and data collection to increase agricultural profitability and sustainability. These technologies aid in efficient resource utilization, conservation of biodiversity, and offering equitable access to farm-related technologies. However, they also face challenges such as high capital investment and need for appropriate policy interventions.

Senthil Kumar et al. [1], utilize Deep Learning and Machine Learning algorithms to address agricultural challenges. The study focuses on crop recommendation systems based on environmental conditions, nutrient requirement analysis, and Growing Degree Days calculations. It also involves weed identification and herbicide recommendation using the Resnet152V2 pre-trained keras model for image classification. Distinguished from the invention, the paper emphasizes disease prediction, market price analysis, and production prediction through a custom-trained model, highlighting a specific focus on disease management and economic analysis in farming.

Cheruku et al. [2], explore digital agriculture technologies in India, assessing their role in sustainable agriculture. The paper discusses the potential and challenges of these technologies in enhancing environmental, economic, and social sustainability. The authors analyze current practices, governmental policies like Digital Agriculture and Precision Agriculture, and identify barriers to technology adoption. They offer policy recommendations to address these challenges, aiming to improve the agriculture sector's sustainability.

Patrick et al. [3], focus on a crop prediction system utilizing machine learning algorithms to enhance agricultural efficiency and productivity. The system uses models trained on geographic and agronomic data to predict crop yields and identify optimal farming operations. The methodology includes analyzing land-specific information such as location, weather, and soil composition. This paper highlights the application of machine learning in enhancing decision-making and cultivation practices for improved crop production.

Alex et al. [4], study on smart agriculture in India examines the use of IoT and machine learning to enhance agricultural practices. Their research, published in Wireless Personal Communications, delves into the challenges of rural network

infrastructure, and proposes innovative solutions, including new architectures and predictive models for crop pricing and irrigation. This work is significant in shaping the future of technology-driven agriculture in India, aiming to improve farmer productivity and decision-making processes.

Barbedo [5], study explores the application of deep learning in plant disease classification, highlighting its potential benefits but also addressing overlooked challenges. It emphasizes the shift towards deep learning since 2015, particularly through Convolutional Neural Networks (CNNs) and transfer learning. The study conducts a detailed analysis of factors affecting deep learning's effectiveness, derived from CNN experiments and misclassification analysis.

Ishana Attri's [6], analysis delves into the transformative impact of deep learning (DL) on agriculture, spanning various domains such as forecasting crop yields, detecting plant stress, identifying weeds and pests, diagnosing diseases, and implementing precision farming techniques. It underscores the prevalence of supervised learning networks like Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Alex Net, and ResNet in driving economic advancement within the sector. Nevertheless, it underscores the imperative of innovating new DL methodologies to bolster model accuracy and diminish inference time for real-world implementations. By pinpointing critical areas of research deficiency, particularly in the realm of novel technique development, the study underscores the ongoing necessity for research endeavors to fully harness DL's potential in precision agriculture, ultimately fostering sustainable agricultural progress.

Eva Cetinic's [7], explores Convolutional Neural Networks (CNN) for art-related image classification tasks using large digitized fine art collections. It conducts extensive fine-tuning experiments across five classification tasks on three datasets, achieving state-of-the-art results. It introduces a novel task of classifying artworks based on their association with a specific national artistic context. The study also investigates the transferability of deep representations across domains, showing that networks pre-trained for scene recognition and sentiment prediction outperform those pre-trained for object recognition. Practical applications are discussed, highlighting the ability of fine-tuned networks to enhance image similarity retrieval for both style and content in online art collections.

Chakraborty et al. [8], highlights how Indian farmers cannot opt for the most suitable crops through conventional methods, which leads to significant socio-economic consequences, including increased movement of people to cities and, in some cases, suicide. In this context, the paper introduces a system that will use machine learning to recommend crops based on the existing factors, such as soil conditions, sowing seasons, and geographical locations. Such a system falls under the umbrella of precision agriculture, which essentially involves site-specific crop management, and has been ever so relevant in the developing world. The solution suggests the use of improved, data-driven, scientific methods as inputs to the process of improving crop choice processes among farmers with a view to increasing agricultural productivity.

Shikha Ujjainia [9], discusses the transformative impact of integrating technology with crop yield prediction methods, emphasizing the role of machine learning in optimizing agricultural outcomes. By combining different algorithms, technological devices can achieve highly accurate predictions with minimal deviation from expected production levels. The study highlights the integration of three separate models into predictive systems, showcasing how information technology interventions, particularly through ensemble algorithms, empower the agricultural industry to maintain expected crop production levels effectively.

Yogesh Gandge [10], addresses the significance of predicting crop yield in India, where agriculture is a crucial source of livelihood and economy but faces challenges like natural calamities leading to financial losses and farmer suicides. Accurate prediction can aid farmers and government agencies in planning storage, pricing, and trade. Given the complexity of factors influencing crop yield, data mining techniques are employed to analyze vast datasets including soil quality and nutrient levels. The study discusses various data mining techniques utilized for crop yield prediction, emphasizing the importance of feature extraction and classifier selection. It summarizes results from different algorithms used by researchers, highlighting their accuracy and recommendations for future applications.

Doshi et al. [11], explore the integration of Big Data Analytics and Machine Learning to support Indian farmers in making precise crop-related decisions through their work by recognizing the traditional reliance on intuition and historical practices in agriculture, the authors propose a system that addresses the complexities of modern agricultural demands. AgroConsultant is designed to guide farmers in selecting appropriate crops based on a multitude of factors including the sowing season, geographical location of farms, soil characteristics, and environmental conditions like temperature and rainfall. This system not only aids in maximizing crop output by adapting to current conditions but also mitigates the risks of poor agricultural decisions, potentially enhancing the agricultural economy stability and efficiency.

Aditya et al. [12], explore the awareness of Minimum Support Price (MSP) among Indian farmers and its influence on crop diversification. Utilizing data from the National Sample Survey Office's 70th round, the authors reveal

that awareness of MSP among rural agricultural households is notably low, with only 23.72% and 20.04% informed about MSP for kharif and rabi crops, respectively. The study employs a probate model to suggest that for MSP to be effective in providing a safety net for farmers, it must be accompanied by efficient procurement processes and increased awareness efforts. Additionally, using the Heckman selection model, the research finds no significant link between MSP awareness and the decision to specialize in certain crops. The paper underscores the necessity for improved informational outreach and policy support to enable farmers to benefit from MSP and possibly influence crop diversification decisions.

Mahesh et al. [13], examine the impact of various government schemes on India's agricultural sector with the goal of promoting financial inclusion and enhancing farmers' welfare. The research evaluates the performance of schemes such as MSP, PMKSY, PMFBY, e-NAM, and PM-KISAN, among others, in terms of increasing productivity, profitability, and supporting farmers' transition to modern agricultural practices. These initiatives, coupled with technology adoption and support from Self-Help Groups (SHGs), are highlighted for their role in improving food supply chains, GDP contribution, and rural employment. Additionally, innovative efforts like Kisan Rail and Krishi Udaan are recognized for their potential to double farmers' incomes, improve irrigation, ensure crop insurance, and facilitate direct subsidy transfers. The study underscores the transformative effect of these schemes on the agricultural economy and the importance of connecting beneficiaries to the benefits, aiming for a holistic development of the country's economy.

Yadav et al. [14], investigates farmers' awareness and perceptions regarding soil testing practices, emphasizing its crucial role in optimizing chemical fertilizer use for enhanced crop production and profitability. Findings indicate a broad awareness among farmers about soil testing, primarily gained through interactions with Krishi Vigyan Kendra scientists and State Department of Agriculture personnel. Despite recognizing the importance of soil testing for improved crop yield, a significant portion of respondents displayed a reluctance towards adopting these practices, often holding unfavorable attitudes. The study recommends that efforts by Krishi Vigyan Kendra and the Department of Agriculture to promote soil testing through targeted training and awareness campaigns could enhance adoption rates, ultimately leading to better farming outcomes.

Raja et al. [15], explore the underutilization of agriculture in India's GDP despite it employing about half of the country's population. They propose a predictive model that aims to address the gap between agricultural employment and its contribution to the GDP, which is only 14%. Their model uses a sliding window non-linear regression technique to analyze factors such as rainfall, temperature, market prices, land area, and past crop yields. The study is conducted across several districts in Tamil Nadu, India, providing tailored crop recommendations. This model not only aims to optimize crop yields and profitability but also addresses the broader socio-economic challenges faced by farmers.

Gawade et al. [16], address the challenges faced by Indian farmers, whose agricultural productivity is influenced by various economic, environmental, and habitat-related factors. The authors propose a tool that leverages machine learning and artificial intelligence technologies to improve crop yields across different geographical regions. The tool gathers data directly from farmers and provides tailored recommendations, which include soil characteristics and optimal cultivation practices. It suggests the most suitable fertilizers and crops to enhance yield based on the specific conditions of the farmer's land. This project utilizes advanced algorithms such as machine learning, regression, and recommendation systems to ensure accuracy and effectiveness. The data collected is stored for future use, enhancing the tool's utility over time. This approach aims to mitigate yield losses and support sustainable farming practices.

Ibrahim et al. [17], give an exhaustive overview of the application of machine learning algorithms in disease diagnostics. They focus on using a variety of algorithms such as Naïve Bayes, logistic regression, support vector machines, K-nearest neighbors, K-means clustering, decision trees, and random forests for the prediction and diagnosis of diseases. These techniques that analyze a wide array of medical data are responsible for facilitating the early prediction of diseases, which is quite crucial for effective treatment interventions. The authors, in a very critical manner, evaluate the performance of these algorithms, pointing out that these algorithms can transform healthcare diagnostics and are greatly responsible for enhancing patient outcomes by implementing timely medical intervention.

In [20], presents an innovative approach to smart farming by integrating artificial intelligence techniques in the selection of manure composition. The authors propose a model that utilizes advanced algorithms to optimize manure use for sustainable farming practices. The study highlights the significance of AI in enhancing agricultural productivity while ensuring environmental sustainability. The findings demonstrate the potential of AI to improve farm management by providing data-driven insights into resource allocation, which is crucial for achieving more efficient and eco-friendly farming practices. This paper offers valuable contributions to the growing body of literature on smart farming and AI applications, and its relevance aligns with the ongoing advancements in sustainable agriculture.

Therefore, its inclusion is essential for updating the literature review and reinforcing the contemporary context of the current study.

The literature on digital agriculture technologies highlights the transformative role of precision agriculture, machine learning, and deep learning in addressing agricultural challenges. Prior research emphasizes tools for crop recommendation, disease prediction, market analysis, and resource optimization, leveraging models like CNNs, ResNet, and ensemble algorithms. However, challenges such as high costs, limited infrastructure, and low awareness persist. Distinct from existing work, the proposed system integrates a multilingual platform powered by Generative AI to offer a holistic solution, combining government scheme guidance, AI-driven disease detection, location-based soil testing, and MSP center identification. By unifying these features into an accessible and cost-effective platform, it bridges the gap between traditional and advanced techniques, addressing gaps in inclusivity, accessibility, and comprehensive decision-making while building on the strengths of earlier studies.

## 3 Design and implementation

This work introduces six modules, as illustrated in Figs. 1 and 2, ie; crop recommendation, disease identification, LLM-powered scheme advisor, crop planning tool, soil testing station, and nearby MSP centers. Figure 1 depicts the diagram illustrates a multilingual web application designed to provide comprehensive agricultural services and support for farmers by leveraging advanced technologies. The platform integrates key features to address critical farming challenges. On the agricultural services side, it includes AI-driven disease detection to identify and provide solutions for plant diseases based on farmer-submitted images, crop recommendation systems that deliver tailored suggestions to optimize yields, and a locator for nearby soil testing centers to enhance soil quality. On the support and information side, the platform offers tools such as an MSP center locator to help farmers find the nearest Minimum Support Price (MSP) centers for fair pricing, an AI-enhanced scheme advisor to guide farmers on government schemes and subsidies, and a crop planning tool powered by Large Language Models (LLMs) to streamline planting schedules and resource management. By integrating these features, the web application bridges the gap between traditional farming methods and modern technology, empowering farmers with tools to improve productivity, sustainability, and access to critical information. The project is implemented as a web application using the Flask framework. Below are detailed descriptions of each module:

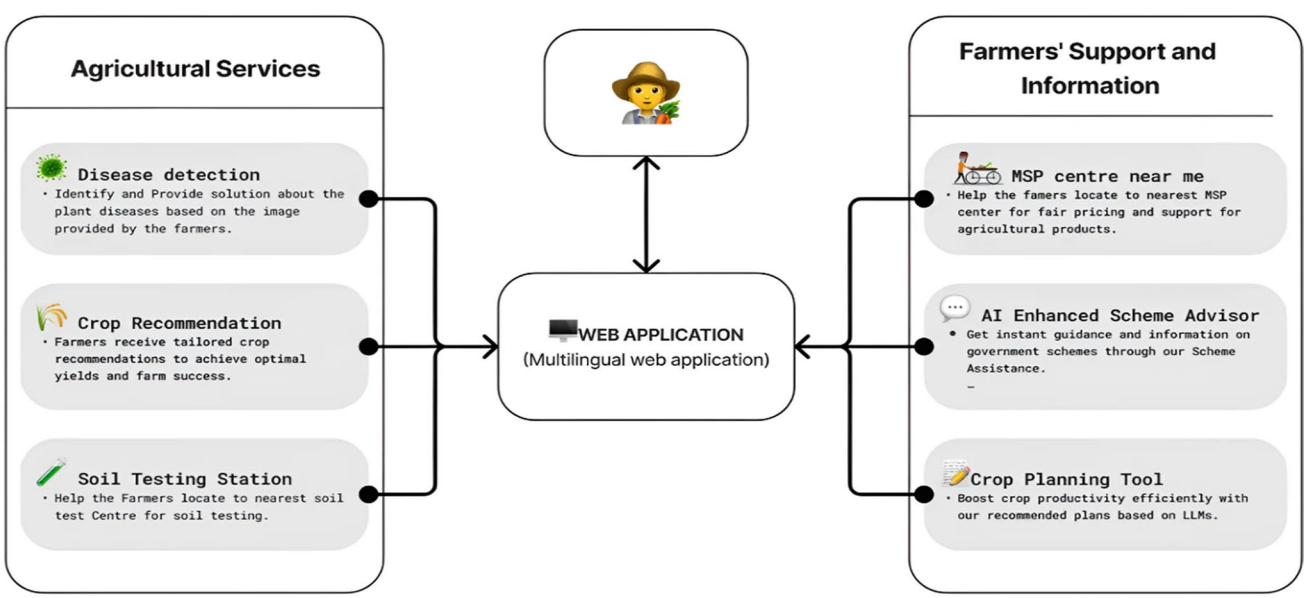



**Fig. 1** System architecture

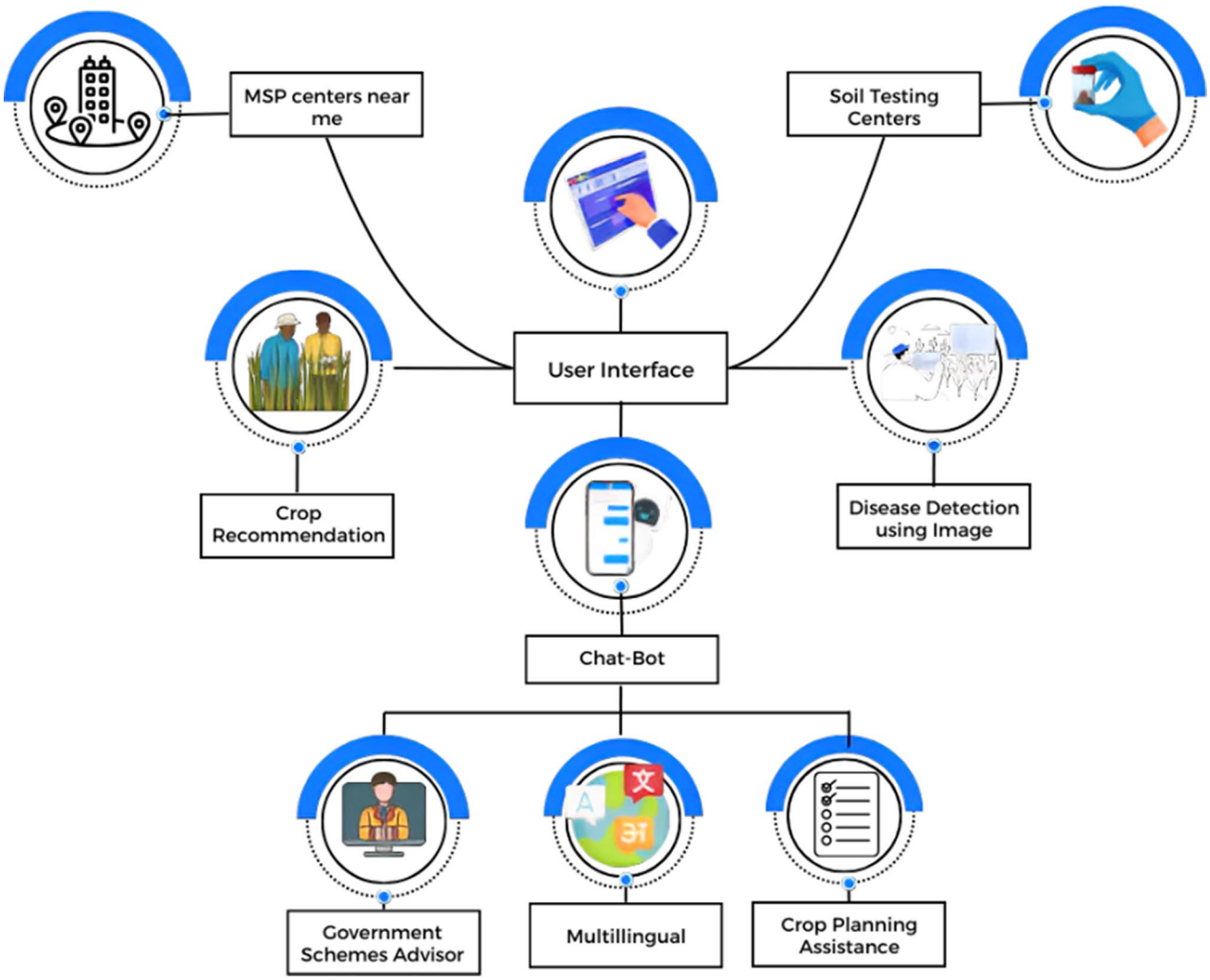



**Fig. 2** Proposed flow of model

## 3.1 Crop recommendation

### 3.1.1 Datasets used

The Crop Recommendation module utilizes a dataset named "Crop_recommender.csv" to provide recommendations on suitable crops for cultivation.

All these datasets were obtained from Kaggle website [“https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset”]. Figure 3 provides a summary of the datasets utilized in this module.

The Crop Recommendation module utilizes a dataset for model training, which includes essential attributes such as temperature, humidity, average rainfall, soil pH, nitrogen requirement ratio, potassium requirement ratio, and phosphorous requirement ratio. Upon importing the dataset, it undergoes a check for missing values per attribute, revealing that all attributes in the Crop recommendation dataset are complete. Subsequently, the data types of the attributes are identified, as illustrated in Fig. 4. Furthermore, Fig. 5 illustrates the distribution of crop instances in the dataset, indicating a uniform distribution across all crops. The steps involved in this module are outlined as follows:

**3.1.1.1 Step 1: importing libraries and dataset** To employ ML algorithms and preprocessing tools effectively, it is necessary to import specific libraries. These libraries enable streamlined model development and prediction. Com-

**Fig. 3** Crop recommendation dataset sample values

| | N | P | K | temperature | humidity | ph | rainfall | label |
|---|---|---|---|---|---|---|---|---|
| 0 | 90 | 42 | 43 | 20.879744 | 82.002744 | 6.502985 | 202.935536 | rice |
| 1 | 85 | 58 | 41 | 21.770462 | 80.319644 | 7.038096 | 226.655537 | rice |
| 2 | 60 | 55 | 44 | 23.004459 | 82.320763 | 7.840207 | 263.964248 | rice |
| 3 | 74 | 35 | 40 | 26.491096 | 80.158363 | 6.980401 | 242.864034 | rice |
| 4 | 78 | 42 | 42 | 20.130175 | 81.604873 | 7.628473 | 262.717340 | rice |
| ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 2195 | 107 | 34 | 32 | 26.774637 | 66.413269 | 6.780064 | 177.774507 | coffee |
| 2196 | 99 | 15 | 27 | 27.417112 | 56.636362 | 6.086922 | 127.924610 | coffee |
| 2197 | 118 | 33 | 30 | 24.131797 | 67.225123 | 6.362608 | 173.322839 | coffee |
| 2198 | 117 | 32 | 34 | 26.272418 | 52.127394 | 6.758793 | 127.175293 | coffee |
| 2199 | 104 | 18 | 30 | 23.603016 | 60.396475 | 6.779833 | 140.937041 | coffee |

**Fig. 4** Attributes in the data

```
N               int64
P               int64
K               int64
temperature   float64
humidity      float64
ph            float64
rainfall      float64
label          object
dtype: object
```

monly used libraries include NumPy, pandas, matplotlib, seaborn, and sklearn. Before applying any machine learning algorithm, it is essential to preprocess the dataset. This involves handling missing values, encoding categorical variables, scaling numerical features, and splitting the dataset into training and testing sets. For instance, missing values are imputed using mean values, and numerical features were normalized to improve model performance. For agricultural applications, each crop is associated with various environmental factors such as temperature, humidity, rainfall, and soil nutrient levels including Nitrogen, Potassium, and Phosphorus. In this particular study, the Random Forest Classifier model was utilized [19]. The dataset used for training and testing the models is called the crop recommendation dataset. Figure 3 offers a sneak peek into the dataset utilized for crop recommendation.

**3.1.1.2 Step 2: descriptive analysis** To obtain a best predictive model descriptive analytics is to be performed in prior. Descriptive analytics provides an idea of how the dataset looks like and helps to draw new insights.

Once identifying that there are no missing values, in Fig. 6 the datatype of the attributes is identified.

**Fig. 5** Crop classes

```
rice            100
maize           100
jute            100
cotton          100
coconut         100
papaya          100
orange          100
apple           100
muskmelon       100
watermelon      100
grapes          100
mango           100
banana          100
pomegranate     100
lentil          100
blackgram       100
mungbean        100
mothbeans       100
pigeonpeas      100
kidneybeans     100
chickpea        100
coffee          100
```

**Fig. 6** Crops in dataset & datatypes

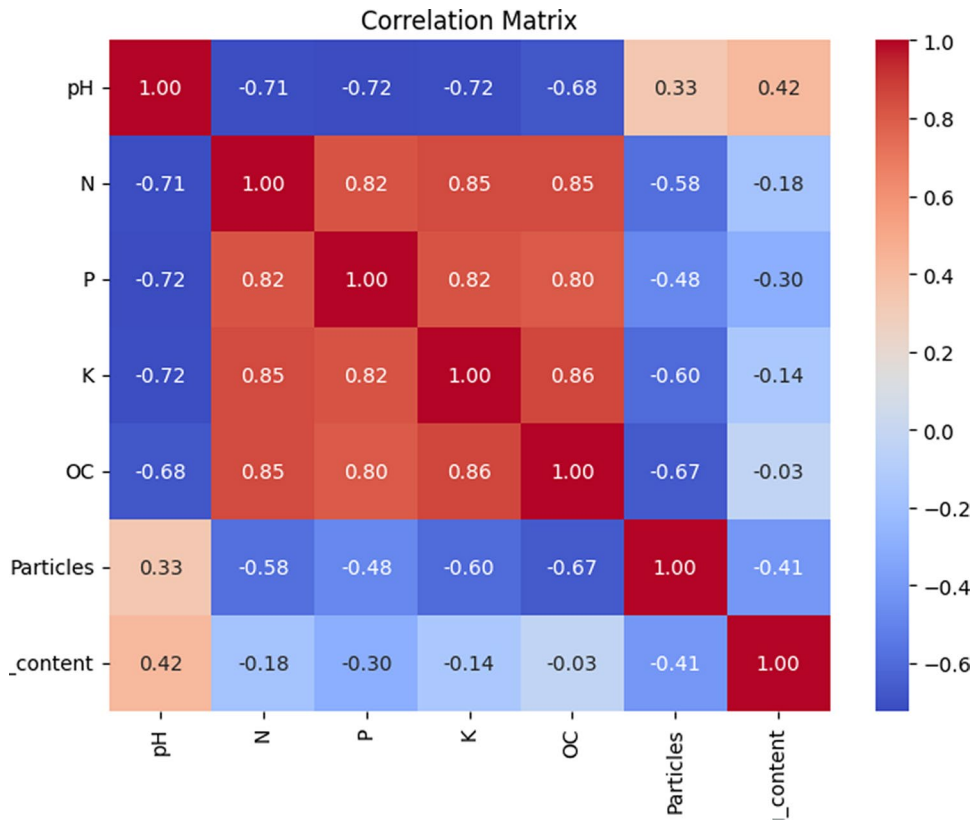
```
Data columns (total 8 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   N            2200 non-null   int64
 1   P            2200 non-null   int64
 2   K            2200 non-null   int64
 3   temperature  2200 non-null   float64
 4   humidity     2200 non-null   float64
 5   ph           2200 non-null   float64
 6   rainfall     2200 non-null   float64
 7   label        2200 non-null   object
dtypes: float64(4), int64(3), object(1)
memory usage: 137.6+ KB
```

**3.1.1.3 Step 3: data visualization** After obtaining the fundamental details about the dataset, data visualization is conducted to analyze the dataset visually. A correlation matrix, also known as a relationship lattice, presents correlation coefficients between attributes in a tabular format.

Here, the attributes are addressed in the first line, and in the first column. Figure 7 shows the correlation matrix for the dataset.

**3.1.1.4 Step 4: splitting the data into train and test sets** After identifying outliers and visualizing the attributes, the initial step in constructing the model involves dividing the dataset into training and testing sets. Training data, essential for

**Fig. 7** Correlation matrix




training machine learning algorithms, serves as the initial input for these algorithms to learn and make accurate predictions or perform desired tasks. The testing set, on the other hand, is a separate dataset used to evaluate the performance of the finalized model trained on the training data. The split ratio between the training and testing sets is 80:20.

**3.1.1.5 Step 5: model building** The objective of this study is to classify crops using Machine Learning Classification algorithms. The steps involved in building the model are outlined as follows:

- Importing the model from the library.
- Defining the model.
- Fitting the training and testing data into the model.
- Testing the trained model on the testing dataset.

Figure 8 presents the outcomes obtained from the Random Forest Classifier. The predicted values for the testing set, indicated as y_predictions, are presented as a list. Subsequently, the accuracy scores for both the training and testing sets are provided, along with the overall accuracy score of the model for the crop recommendation dataset.

## 3.2 Disease detection

### 3.2.1 Step 1: dataset

The dataset used for this module is Plant Village dataset (see Fig. 9). PlantVillage dataset is a collection of agricultural images and related information used for research and development in the field of plant pathology and crop health.

### 3.2.2 Step 2: splitting the training, testing set, and validation set

The dataset is divided into three segments: training, testing, and validation data. Validation data is utilized during the execution of epochs. The training set comprises 2701 images, while the testing set consists of 1447 images. Additionally,

**Fig. 8** Model building results

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| apple | 1.00 | 1.00 | 1.00 | 22 |
| banana | 1.00 | 1.00 | 1.00 | 17 |
| blackgram | 1.00 | 0.95 | 0.98 | 22 |
| chickpea | 1.00 | 1.00 | 1.00 | 19 |
| coconut | 1.00 | 1.00 | 1.00 | 15 |
| coffee | 1.00 | 1.00 | 1.00 | 23 |
| cotton | 1.00 | 1.00 | 1.00 | 16 |
| grapes | 1.00 | 1.00 | 1.00 | 21 |
| jute | 0.83 | 1.00 | 0.90 | 19 |
| kidneybeans | 1.00 | 1.00 | 1.00 | 26 |
| lentil | 1.00 | 1.00 | 1.00 | 19 |
| maize | 0.94 | 1.00 | 0.97 | 17 |
| mango | 1.00 | 1.00 | 1.00 | 18 |
| mothbeans | 1.00 | 1.00 | 1.00 | 24 |
| mungbean | 1.00 | 1.00 | 1.00 | 23 |
| muskmelon | 1.00 | 1.00 | 1.00 | 19 |
| orange | 1.00 | 1.00 | 1.00 | 18 |
| papaya | 1.00 | 1.00 | 1.00 | 16 |
| pigeonpeas | 1.00 | 1.00 | 1.00 | 21 |
| pomegranate | 1.00 | 1.00 | 1.00 | 25 |
| rice | 1.00 | 0.80 | 0.89 | 20 |
| watermelon | 1.00 | 1.00 | 1.00 | 20 |
| accuracy | | | 0.99 | 440 |
| macro avg | 0.99 | 0.99 | 0.99 | 440 |
| weighted avg | 0.99 | 0.99 | 0.99 | 440 |

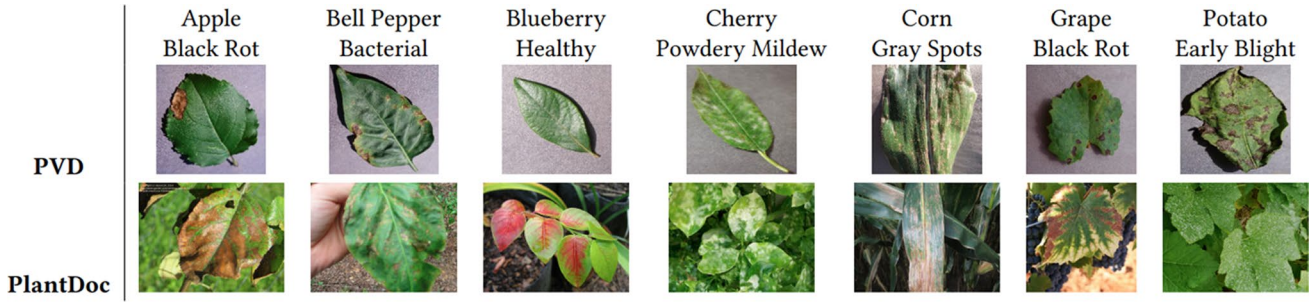



**Fig. 9** Plant disease dataset

the validation set includes 675 images. Data augmentation techniques were employed to augment the number of training images, involving rotations, flips, and rescaling of the images.

### 3.2.3 Step 3: model architecture

The ResNet152V2 pre-trained Keras model is commonly utilized in most implementations for image classification tasks. In this implementation, a custom neural network has been developed to classify images. Deep Residual Networks are networks with convolutional, pooled, activating, and fully connected layers mounted on top of each other. In the proposed approach, a Convolutional Neural Network (CNN) architecture was established, comprising of 5 Conv2D layers. Following these convolutional layers, Global Average Pooling2D was applied, subsequently followed by Dropout layers to mitigate overfitting. Dropout layers were strategically placed in between each hidden layer to alleviate overfitting by removing unnecessary neurons and refining the model's parameter selection process for image identification tasks. The

output layer comprised a single neuron with the SoftMax activation function for classification purposes. This module executed 25 epochs with a batch size of 32. The epochs were trained using the training data and validated using the validation data. Additionally, checkpoints were defined and stored to facilitate future weight updates. Figure 10 shows the Disease Detection model Architecture.

## 3.3 LLM powered scheme advisor

The development of LLM powered Scheme Advisor for the agricultural sector represents a significant stride in leveraging artificial intelligence to provide targeted, real-time advice on government schemes. Central to this innovation is the fine-tuned Mistrial-7B model, which, together with a custom-developed dataset for Indian government schemes for Agriculture, can form the backbone of a system designed to enhance decision-making for agricultural stakeholders.

### 3.3.1 Step 1: integration with the Mistrial-7B model

The Mistrial-7B model forms the core of the system, meticulously refined with a specialized dataset that includes comprehensive details on contemporary agricultural government schemes. This refinement process is key to ensuring the model's effectiveness in interpreting user queries and generating accurate, relevant responses.

### 3.3.2 Step 2: custom dataset development

The dataset used in the model, which includes around 350 Q&A and 33 agricultural initiatives, is derived from reliable websites like the Ministry of Agriculture and Farmers Welfare website of the Government of India. This guarantees that farmers obtain timely and accurate recommendations relevant to India's agricultural landscape https://github.com/AnKiTu03/Krishi-chatbot

### 3.3.3 Step 3: response generation and delivery

Using specific knowledge from our dataset, the Mistrial-7B model is excellent at creating accurate and useful responses. This makes it easier to provide professional guidance on utilizing government programs for the agriculture industry.

## 3.4 Crop Planning Tool

The Crop Planning Tool prioritizes a farmer-centric design, featuring an intuitive user interface. Designed to make planning planting schedules and managing resources easier, the interface provides a user-friendly experience, encouraging efficient crop management practices.

### 3.4.1 Step 1: API integration

The system makes use of the ChatGPT-4 model for sending user queries and receiving responses. API calls are implemented using appropriate programming languages, ensuring secure communication with the ChatGPT-4 model servers.

### 3.4.2 Step 2: data retrieval

For specific queries regarding government schemes, the advisor module interacts with external databases through ChatGPT-4 model to retrieve the latest and relevant information.

### 3.4.3 Step 3: response generation

The system generates coherent responses, incorporating information from both ChatGPT-4 model, providing users with accurate guidance on government schemes.

**Fig. 10** Disease detection model architecture

```
Model: "sequential_1"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d (Conv2D)             (None, 100, 100, 32)      896

 activation (Activation)     (None, 100, 100, 32)      0

 batch_normalization (Batch  (None, 100, 100, 32)      128
 Normalization)

 max_pooling2d (MaxPooling2  (None, 33, 33, 32)        0
 D)

 dropout (Dropout)           (None, 33, 33, 32)        0

 conv2d_1 (Conv2D)           (None, 33, 33, 64)        18496

 activation_1 (Activation)   (None, 33, 33, 64)        0

 batch_normalization_1 (Bat  (None, 33, 33, 64)        256
 chNormalization)

 conv2d_2 (Conv2D)           (None, 33, 33, 64)        36928

 activation_2 (Activation)   (None, 33, 33, 64)        0

 batch_normalization_2 (Bat  (None, 33, 33, 64)        256
 chNormalization)

 conv2d_3 (Conv2D)           (None, 33, 33, 64)        36928

 activation_3 (Activation)   (None, 33, 33, 64)        0

 batch_normalization_3 (Bat  (None, 33, 33, 64)        256
 chNormalization)

 max_pooling2d_1 (MaxPoolin  (None, 16, 16, 64)        0
 g2D)

 dropout_1 (Dropout)         (None, 16, 16, 64)        0

 conv2d_4 (Conv2D)           (None, 16, 16, 128)       73856

 activation_4 (Activation)   (None, 16, 16, 128)       0

 batch_normalization_4 (Bat  (None, 16, 16, 128)       512
 chNormalization)

 conv2d_5 (Conv2D)           (None, 16, 16, 128)       147584

 activation_5 (Activation)   (None, 16, 16, 128)       0

 batch_normalization_5 (Bat  (None, 16, 16, 128)       512
 chNormalization)

 max_pooling2d_2 (MaxPoolin  (None, 8, 8, 128)         0
 g2D)

 dropout_2 (Dropout)         (None, 8, 8, 128)         0

 dropout_2 (Dropout)         (None, 8, 8, 128)         0

 flatten (Flatten)           (None, 8192)              0

 dense (Dense)               (None, 1024)              8389632

 activation_6 (Activation)   (None, 1024)              0

 batch_normalization_6 (Bat  (None, 1024)              4096
 chNormalization)

 dropout_3 (Dropout)         (None, 1024)              0

 dense_1 (Dense)             (None, 8)                 8200

 activation_7 (Activation)   (None, 8)                 0

=================================================================
Total params: 8718536 (33.26 MB)
Trainable params: 8715528 (33.25 MB)
Non-trainable params: 3008 (11.75 KB)
_________________________________________________________________
```

### 3.5 Soil testing centre

This system a comprehensive system designed to provide user utility to search and retrieve nearest soil testing centers based on Users current location.

#### 3.5.1 Step 1: data retrieval

The system makes use of the Selenium for scraping ["https://www.justdial.com/Soil-Testing-Laboratory-near-me"] which gives results based on the users" location and scraped responses includes name and location of the nearest soil testing centers.

#### 3.5.2 Step 2: response generation and delivery

The system generates coherent responses, incorporating information from scraped Data, providing users with accurate guidance on nearest Soil Testing Centers.

### 3.6 MSP centers near me

This system a comprehensive system designed to provide user utility to search and retrieve nearest Crop Selling centers based on Users current location.

#### 3.6.1 Step 1: data retrieval

A dataset containing the names of MSP centers along with their latitude and longitude coordinates has been prepared. Using the prepared dataset, a KD Tree has been constructed. This data structure facilitates efficient querying of MSP centers based on proximity to the user's location.

#### 3.6.2 Step 2: response generation

By utilizing the KD Tree, the system can swiftly identify and retrieve the MSP centers nearest to the user's location.

## 4 Results and discussions

The proposed system offers a web interface that allows users to access the models efficiently. Each section discusses the results of the respective model.

### 4.1 Crop recommendation

The accuracy score of 0.995 has been achieved by our RandomForest Classifier model as shown in Fig. 11.

Figure 12 illustrates the confusion matrix generated for the random forest classifier. The diagonal of the confusion matrix indicates the number of correct predictions.

#### 4.1.1 Frontend

The Fig. 13, presents the Crop Recommendation system, wherein users are required to input information regarding average rainfall, pH levels, and other relevant parameters.

Utilizing the provided parameters, the system predicts the appropriate crop. As depicted in Fig. 13, the user has inputted the values. Upon clicking "predict," the page redirects to the results page, which displays the predicted crop along with its corresponding image.

**Fig. 11** Accuracy score

```
Accuracy: 0.9954545454545455
```

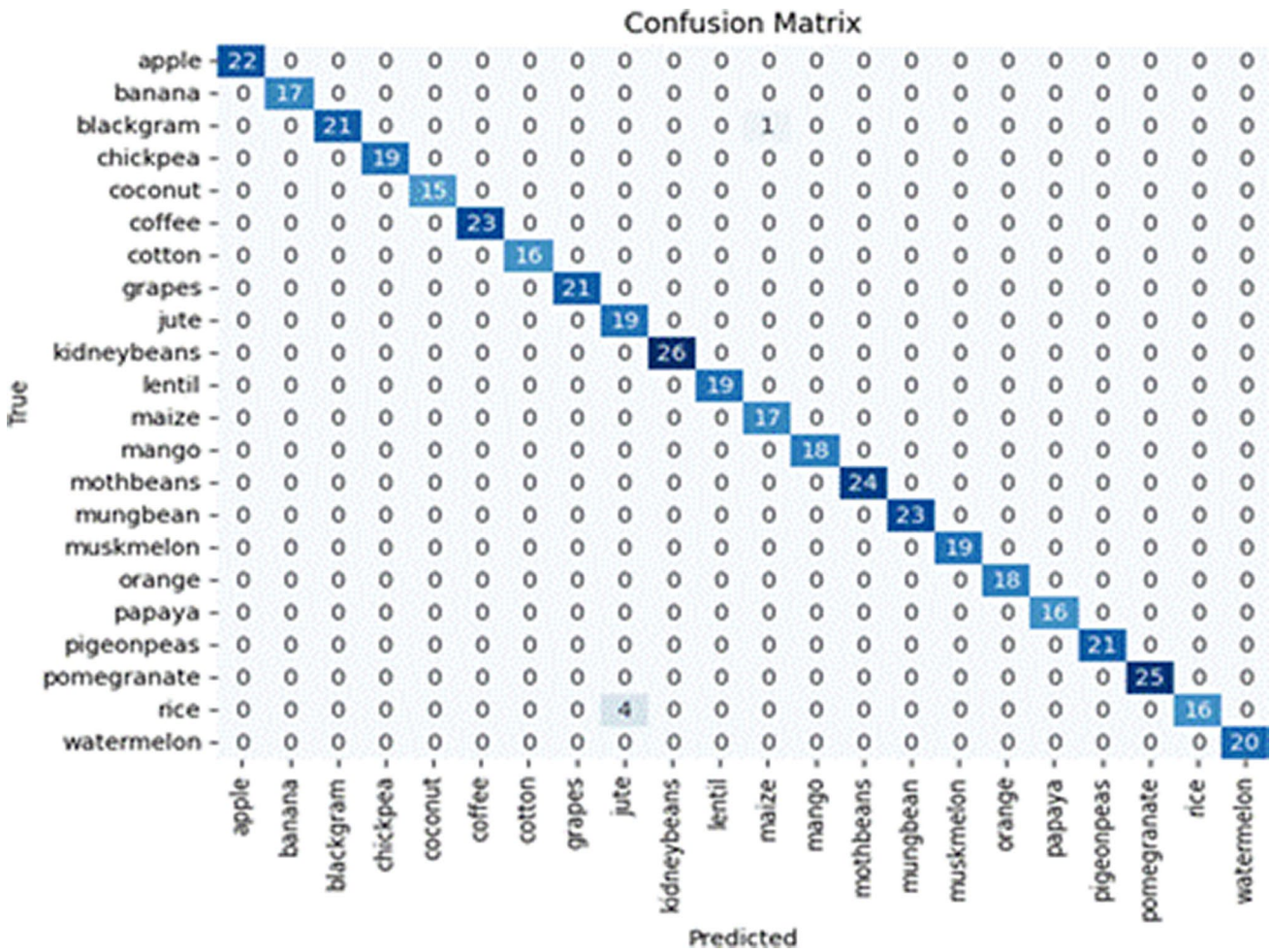



**Fig. 12** Confusion matrix

## 4.2 Disease detection

This module helps in identifying the disease present in crop.

To predict the disease, a custom neural network was trained, achieving an accuracy of 0.915, as indicated in Fig. 14.

A dataframe containing file paths is shuffled and subsequently divided into training and test sets for model development and prediction purposes. Following fine-tuning of the base model and incorporation of additional layers, epochs are executed to train the model using the training data and validate it against the validation data. As depicted in Fig. 15, 30 epochs resulted in a training accuracy of 91.5%.

### 4.2.1 Frontend

To identify the disease, the user must upload an image captured from the field.

Once the user uploads the image of the plant/crop found on the field the system predicts the disease. [as shown in Fig. 16].

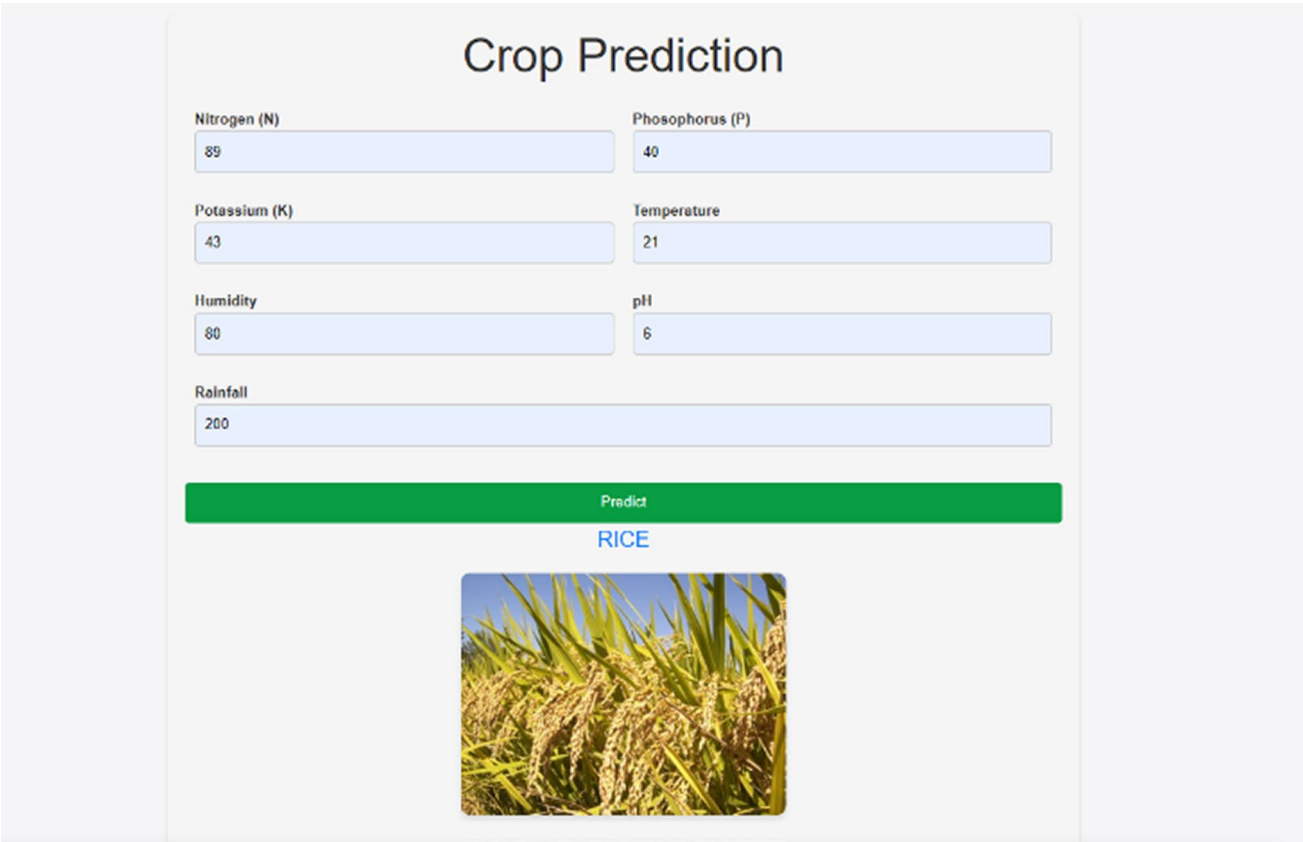



**Fig. 13** Crop recommendation dashboard

```
[INFO] Calculating model accuracy
9/9 [==============================] - 0s 10ms/step - loss: 0.0746 - accuracy: 0.9158
Test Accuracy: 91.57894849777222
```

**Fig. 14** Accuracy

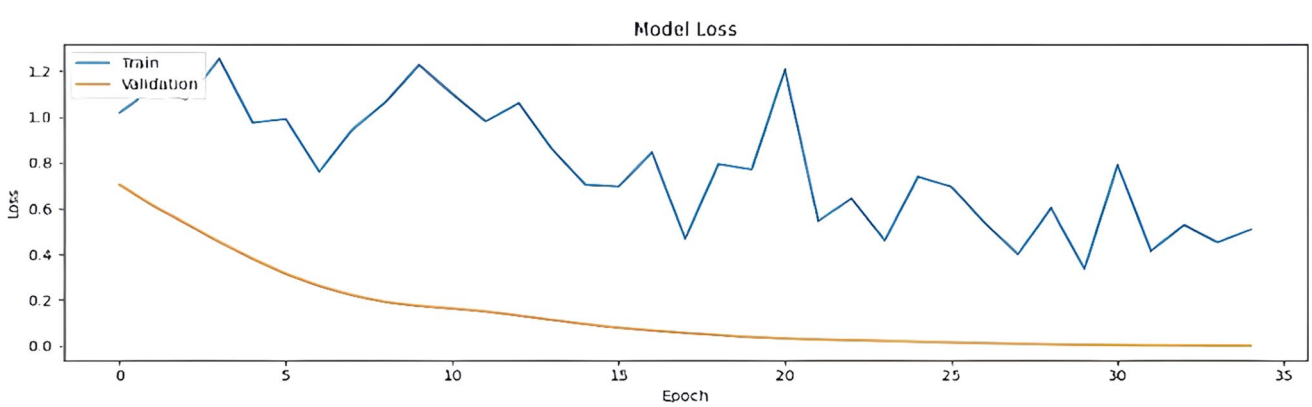



**Fig. 15** Curve

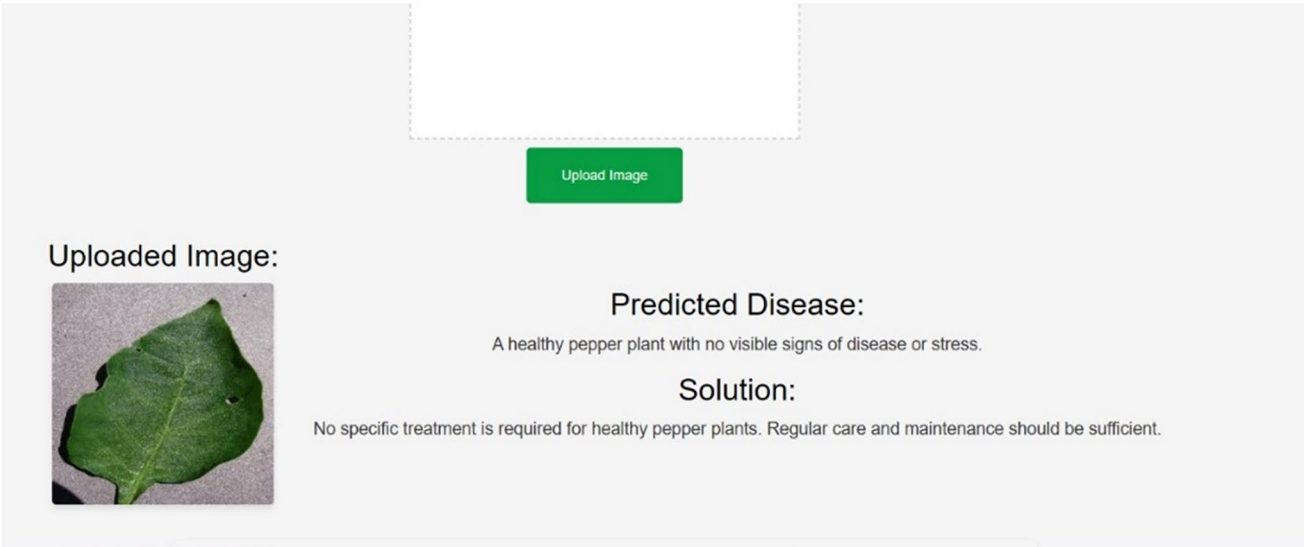



**Fig. 16** Crop disease detection

## 4.3 AI-assisted government scheme advisor

The Chatbot response speed is fast ranging from few seconds to some minutes with appropriate response as per the input provided by the User.

Illustrating the workflow of the AI model fine-tuning process [Fig. 17], showcasing the sequence from the initial input prompt to the final output answer through various stages including preprocessing, the language model, and postprocessing components.

Figure 18 shows a training graph where the model's training loss decreases to approximately 1.28, while the test loss levels of around 1.92, across several evaluation checkpoints, reflecting the model's learning trajectory and potential areas for optimization in loss reduction.

### 4.3.1 Frontend

The following figure shows [Figs. 19 and 20] Government Scheme Advisor chatbot dashboard. Based on the parameters passed, the system responds with the suitable schemes.

## 4.4 Crop planning tool

Designed to make planning planting schedules and managing resources easier, the interface provides a user-friendly experience, encouraging efficient crop management practices. Tool generates appropriate response as per the input provided by the User along with the Multilingual Support in Hindi and English.

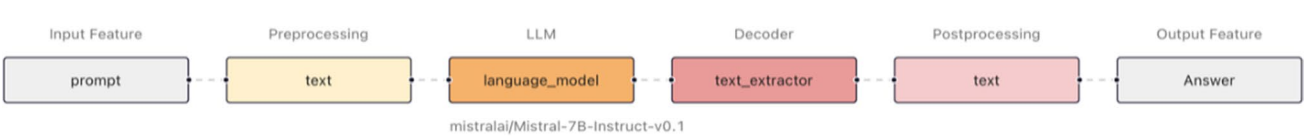



**Fig. 17** Workflow of the AI model fine-tuning process

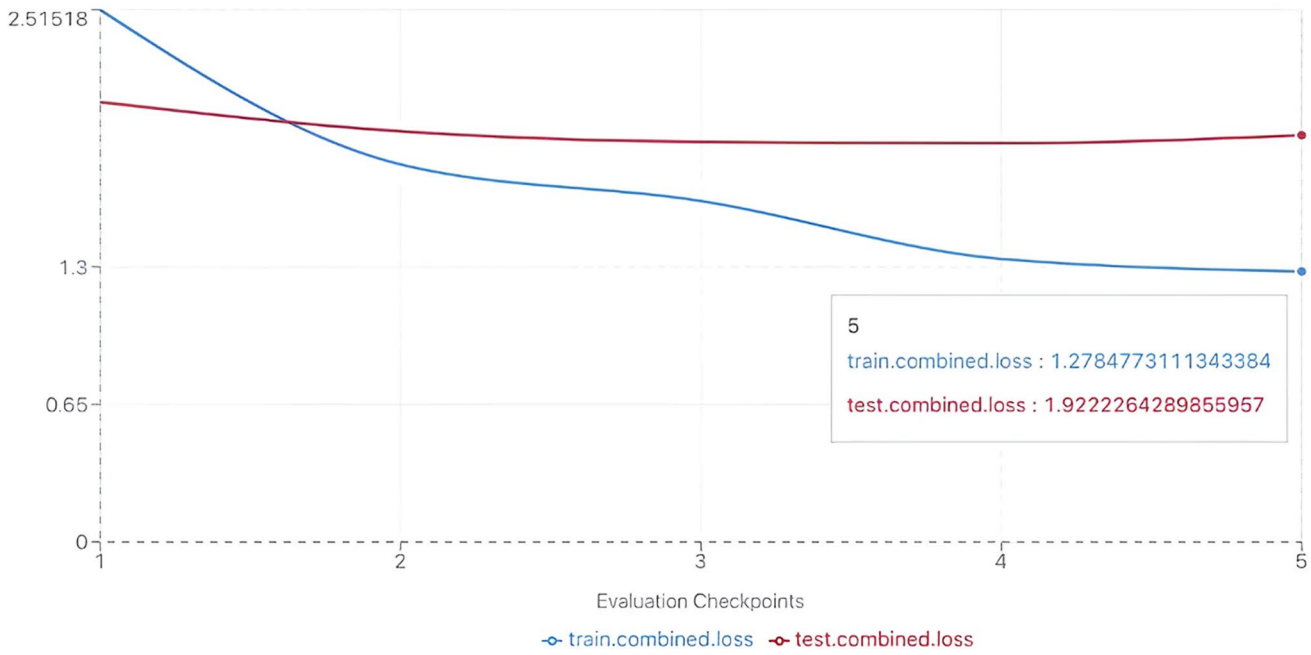



**Fig. 18** Loss Curve

**Fig. 19** Govt Scheme Chatbot

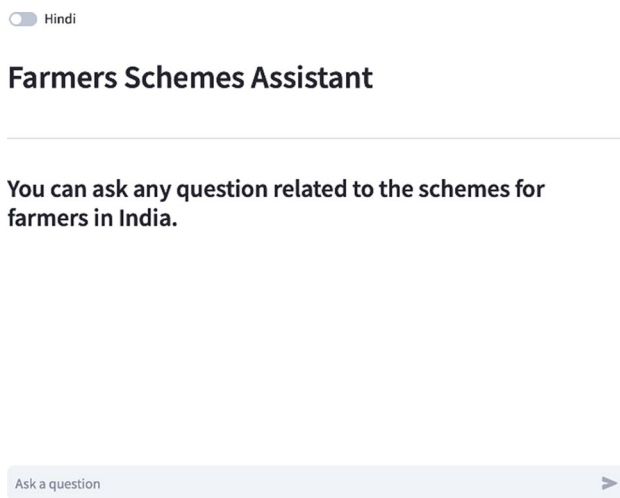



**Fig. 20** Govt Scheme Chatbot

#### 4.4.1 Frontend

The displayed figure exhibits the Crop Planning tool dashboard. Users are prompted to input details regarding investments, start-date, season, crop selection, and available land. Subsequently, the system generates a suitable plan based on the provided parameters.

According to Fig. 21, after the user inputs values and clicks "Enter," the page generates a personalized crop plan based on the provided information.

### 4.5 Soil testing centre

This functionality scrapes data in background for the soil testing centers near users' current location and returns a List of centers on a click of a button.

#### 4.5.1 Frontend

The following figure shows Soil testing Centre functionality.

Upon clicking "Detect," users will receive a list of nearby Soil Testing Centers within seconds (refer to Fig. 22).

### 4.6 MSP near me

By utilizing the KD Tree, the system can swiftly identify and retrieve the MSP centers nearest to the user's location, on a click of a button (Fig. 23).

#### 4.6.1 Frontend

The MSP Center feature allows users to quickly find nearby MSP centers by selecting "Find Center," which displays a list of nearby locations within seconds.

**Fig. 21** Crop planning dashboard

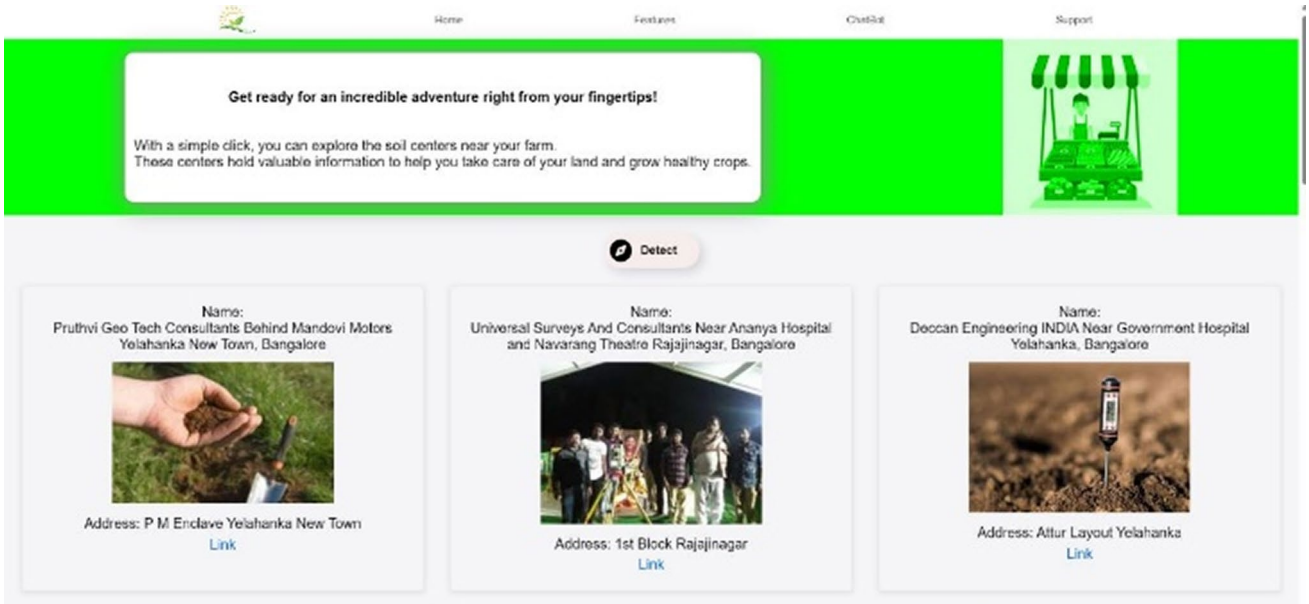



**Fig. 22** Soil testing centers

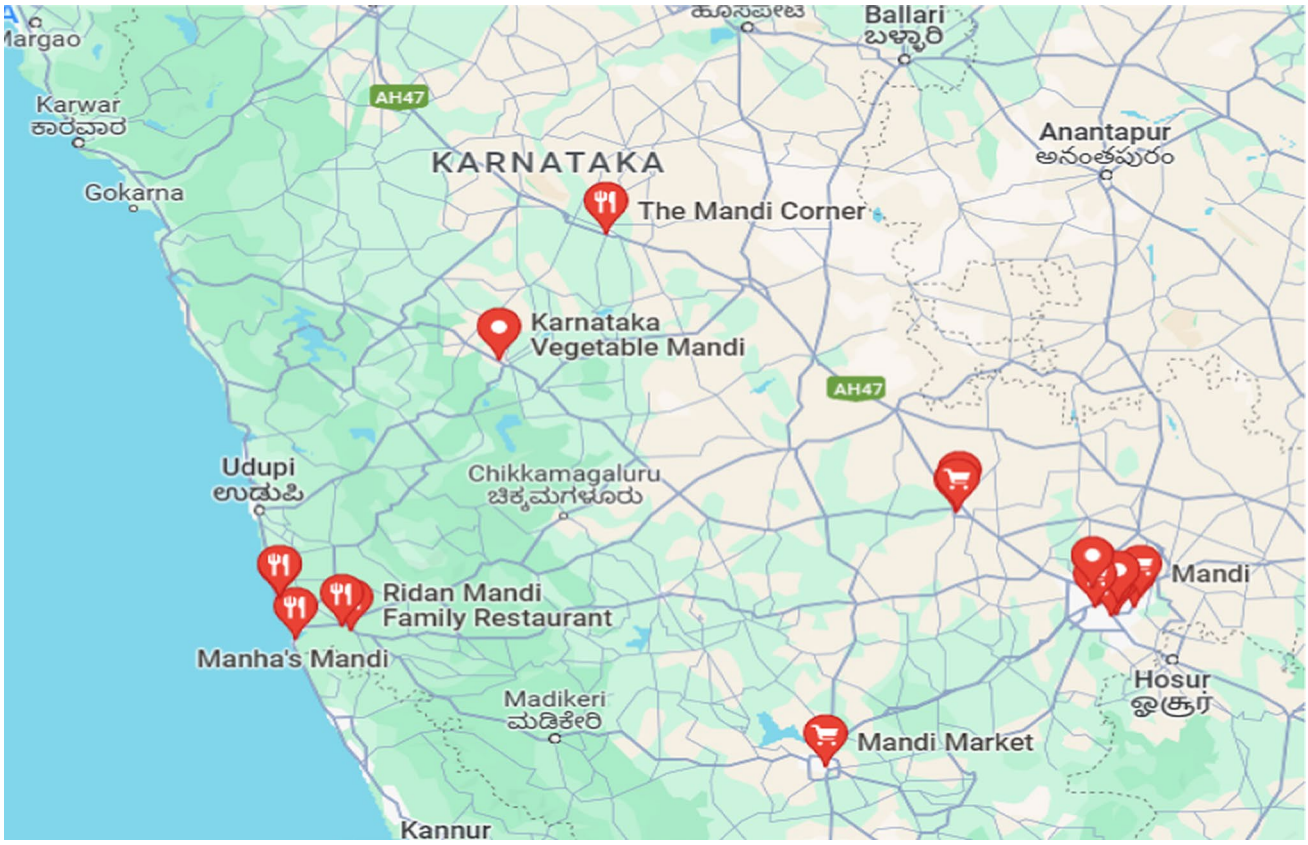



**Fig. 23** MSP Centers near me

## 5 Conclusion

In conclusion, this platform leverages advanced technology to address agricultural challenges effectively by providing tools for government scheme guidance, real-time disease detection, soil testing, personalized crop recommendations, and easy access to Minimum Support Price (MSP) centers. Multilingual support enhances accessibility for farmers from diverse backgrounds, aiming to boost agricultural productivity and sustainability. Future research could focus on enhancing precision agriculture techniques through advanced sensor technologies and data analytics, developing climate resilience strategies for crops, expanding AI applications in crop disease management, improving market intelligence and price forecasting using machine learning, and assessing the socio-economic impacts of technology adoption on farming communities. These efforts aim to cultivate a resilient and prosperous agricultural sector, empowering farmers and preparing the industry for future challenges.

**Author contributions** Sowmya B J, Pradeep Kumar D, Meeradevi HK were identified Initial problem identification, algorithm write-up, analysis, drafting of the manuscript, and simulation. Supreeth S and Rohith S, were responsible for the Literature survey and helped in the initial review process. Ravi kumar B N- was responsible for the Complexity analysis of the research, evaluation of the research work. Divyansh Mishra, Abhishek Koushik, Ankit U Patil were responsible for the figures, final formatting and applied for the journal. All authors worked together to implement and evaluate the integrated system, and approved the final version of the paper.

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**Data availability** The data that support the findings of this study are available from the corresponding author upon reasonable request.

## Declarations

**Competing interests** The authors declare no competing interests.

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